

ROY CARMELLI

FULL STACK · FRONTEND · AI ENGINEERING · CS & NEUROSCIENCE

Israel · Roy.v.carmelli@gmail.com · [LinkedIn](#) · [GitHub/Royc4515](#) · [Portfolio](#) · +972 54-728-7807

SUMMARY

Second-year B.Sc. student in Computer Science & Neuroscience at Bar-Ilan University (graduating 2027) — one of few programs combining software engineering depth with hands-on neuroscience research. Independently ships full-stack apps, Chrome extensions, and AI-powered tools; uses AI-assisted development as a core workflow. Seeking student or junior positions in Full Stack, Frontend, or AI Engineering.

EDUCATION

B.Sc. Computer Science & Neuroscience | Bar-Ilan University | GPA: 86.16

Oct 2024 – 2027

Relevant coursework: Data Structures & Algorithms (93) · OOP in Java (92) · Systems Programming (C/Assembly) · Neuroscience (94) · Biochemistry (99) · Calculus (100) · Linear Algebra (97) · Python Data Processing (98)

PROJECTS

CareerPredict AI | React 19 · Node.js · Express · SQLite · Google OAuth · Render [🔗 live demo](#) [🔗 GitHub](#)

- Full-stack SPA - Google OAuth 2.0 login → multi-step onboarding → AI-generated career portrait via Pollinations.ai (100+ keyword-classified outcomes). React/Vite/Tailwind frontend, Node/Express + Passport.js backend, SQLite with migration logic
- Production-deployed on Render with secure session management, deterministic image seeding per user, and image retry loop (5 attempts, cache-busting)

Wine Sommelier Telegram Bot | Python · Google Gemini 2.5 Flash · Telegram API · Google Sheets · Vercel [🔗 GitHub](#)

- Serverless, event-driven Telegram bot - webhook-triggered Vercel function reads live wine inventory from Google Sheets and delivers food-pairing recommendations via natural language. Webhook token validation and user allowlist for security
- Resilient AI pipeline: Gemini 2.5 Flash with automatic fallback chain (gemini-3.1-flash-lite → gemma-4-31b → gemini-2.5-flash) and exponential backoff. Two-layer memory: active session history (30 msgs, 1-hr TTL) + AI-compressed long-term summary on expiry. 25 unit tests across 5 files with @patch mocking. 34 commits

AI Sidebar - Chrome Extension | JavaScript · Chrome Extensions API · Manifest V3 [🔗 GitHub](#)

- Integrates 6 AI providers (Claude, Gemini, GPT-4o, Grok, Groq, Ollama) via Factory + Strategy + Template Method OOP patterns - BaseProvider abstract class with concrete provider subclasses; ProviderFactory centralizes instantiation. Runtime provider switching with live API key validation
- Service Worker background architecture (MV3) with cross-origin postMessage security, page content extraction (~12,000 char truncation with notice), and custom Markdown-to-HTML renderer

Arkanoid Game | Java 17 · OOP Design Patterns · Custom Geometry Engine [🔗 GitHub](#)

- Java 17 implementation across 20 classes - Observer pattern (HitNotifier/HitListener interfaces), Composite pattern (SpriteCollection, GameEnvironment), and Factory Method (Velocity.fromAngleAndSpeed). Interface-first composition throughout
- Custom geometry engine: 4-case line-segment intersection, epsilon-robust floating-point comparisons, AABB collision detection, 5-zone paddle deflection with angle mapping, and ball-unstuck logic via edge nudging

Personal Portfolio | React 18 · TypeScript · Vite · Tailwind CSS · Framer Motion [🔗 live site](#) [🔗 GitHub](#)

- Deployed portfolio (Vercel, 30 commits) with an embedded endless runner mini-game engine - TypeScript OOP classes (GameEngine, Player, Obstacle, ObstacleManager, SpriteRenderer), AABB collision, physics simulation (gravity/jump/clamp), 7-state player FSM, and sprite sheet animation
- Custom Event Bus (DOM CustomEvents) for zero-prop-drilling component communication. Token-based design system, Framer Motion AnimatePresence transitions, and Tailwind v4 Vite plugin

MILITARY SERVICE

IDF Combat Medic & Medical Coordinator | Gaza Division

Nov 2021 – present

- Operational Leadership:** Directed a specialized medical unit, managing a team that scaled from 12 to 30+ personnel during high-intensity conflict; oversaw complex logistics and personnel deployment under extreme pressure.
- Operations Coordination:** Managed real-time data synchronization and communication protocols within a divisional war room to ensure rapid response and resource optimization across multiple units.
- Crisis Resilience & Decision Making:** Orchestrated high-stakes triage and evacuation logistics during mass-casualty events, demonstrating extreme resilience and technical decision-making clarity in active-threat environments.

SKILLS

Languages: Java 17, Python, C, x86-64 Assembly, JavaScript (ES6+), TypeScript

Web & Full Stack: React, Vite, Node.js, Express, Tailwind CSS, HTML/CSS, Passport.js, SQLite, Google OAuth
Data & Analysis: NumPy, Pandas, Matplotlib, SciPy, Jupyter, pytest
Tools & Platforms: Git/GitHub, Chrome Extensions API (MV3), Render, Vercel, GitHub Pages
AI-Assisted Dev & MCP: Architecture, implementation, debugging, and code review.

LEADERSHIP & ADDITIONAL

StandWithUs Fellowship (2025) - Selected for competitive Israel advocacy and leadership fellowship; trained in public speaking, media engagement, and cross-cultural communication
Languages: Hebrew (native), English (professional proficiency)